Name: Pavan Bhapkar
PRN: 24070126137
Batch: AIML 24-28 B3

Experiment 1

Code:

```
import java.util.Scanner;

public class Calculator {
    double num1, num2;
    public int choice;

    public double addNums(double n1, double n2) {
        return n1 + n2;
    }

    public double multiplyNums(double n1, double n2) {
        return n1 * n2;
    }

    public double subtractNums(double n1, double n2) {
        return n1 - n2;
    }

    public double divideNums(double n1, double n2) {
        return n1 / n2;
    }

    public double modulus(double n1, double n2) {
        return n1 % n2;
```

```
}

public static void main(String[] args) {
    Calculator c = new Calculator();
    Scanner scan = new Scanner(System.in);

    while (true) {
        System.out.println("\nChoose an operation:");
        System.out.println("1. Addition");
        System.out.println("2. Subtraction");
        System.out.println("3. Multiplication");
        System.out.println("4. Division");
        System.out.println("5. Modulus");
        System.out.println("0. Exit");

        c.choice = scan.nextInt();

        if (c.choice == 0) {
            System.out.println("Exiting calculator...");
            break;
        }

        System.out.println("Enter the first number: ");
        c.num1 = scan.nextDouble();
        System.out.println("Enter second number: ");
        c.num2 = scan.nextDouble();

        switch (c.choice) {
            case 1:
```

```
            System.out.println("Sum: " + c.addNums(c.num1, c.num2));
            break;

        case 2:
            System.out.println("Difference: " + c.subtractNums(c.num1, c.num2));
            break;

        case 3:
            System.out.println("Product: " + c.multiplyNums(c.num1, c.num2));
            break;

        case 4:
            if (c.num2 == 0) {
                System.out.println("Cannot divide by zero!");
            } else {
                System.out.println("Division: " + c.divideNums(c.num1, c.num2));
            }
            break;

        case 5:
            System.out.println("Modulus: " + c.modulus(c.num1, c.num2));
            break;

        default:
            System.out.println("Invalid Operation");
    }
}

scan.close();
```

```
    }
}
```

Sample Output

```
Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
0. Exit
1
Enter the first number:
2
Enter second number:
5
Sum: 7.0

Choose an operation:
1. Addition
```

```
Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
0. Exit
7
Enter the first number:
1
Enter second number:
2
Invalid Operation
```

```
Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
0. Exit
0
Exiting calculator...

=== Code Execution Successful ===
```